Last Time:

- Local (Linear) Stability
- Stability of spinning rigid body

Today:

- Attitude Representations

Attitude Representations:

- So far, we've focused on attitude dynamics (integrating Euler's equation to get $\omega(t)$):

$$J\dot{\omega} + \omega \times J\omega = \tau$$

$\dot{\omega} \longrightarrow \omega(t)$

- We're still missing the next part of the puzzle: going from $\omega(t)$ to the full orientation/pose/configuration of the body.

- We've seen one version of this already for rotation matrices:

$$\dot{Q} = Q\,S(\omega)$$

"rotation kinematics" = mapping from $\omega \rightarrow \dot{Q}$

skew-symmetric "cross-product matrix"

* Transition to Matlab Q-integration code. Show how numerical drift destroys $Q^TQ = I$ and we end up with something that's no longer a valid rotation.

You can also ask/tease how you might fix this with SVD (too expensive to run every sim step)

* Definition of "attitude":

Rotation between a body-fixed frame and an inertial frame:

$$^{N}X = {}^{N}Q^{B}\,{}^{B}X$$

* Ways we can parameterize Attitude:

- Euler Angles (roll-pitch-yaw):

pro: Minimal (3 numbers), intuitive

con: Singularities (at 90°), trig functions in kinematics

- Rotation Matrix:

pro: Non-singular, easy to rotate vectors, linear kinematics

con: Redundant! 9 numbers for 3 DOF ⇒ 6 constraints

- Quaternions:

pro: Non-singular, easy to simulate dynamics, linear kinematics

con: Redundant! 4 numbers for 3 DOF ⇒ 1 constraint

- Axis-Angle Vector

pro: Minimal, intuitive

con: Singularity in kinematics at 180°

- Gibbs / Rodrigues Vector / Parameters

  pro: Minimal, polynomial kinematics

  con: Singularities at 180° (MRP at 360°)

A Little Group Theory:

- For practical purposes, a group is a set of matrices (can have other equivalent representations) with the following:

  1) An identity element
  2) An inverse
  3) Closed under multiplication

     For any A and B in the group
     $\Rightarrow AB = C$ must also be in the group

- Examples:

  - Positive reals
  - Discrete symmetry groups of polygons e.g. square D_4 
  - $N \times N$ Invertable Matrices $GL(N)$
  - Rotations $SO(N)$
  - Rigid body motion $SE(3)$

- Continuous groups are called Lie Groups

* The group of 3D rotations is called $SO(3)$

  S = "special" $\Rightarrow \det(Q) = 1$

  O = "orthogonal" $\Rightarrow Q^T = Q^{-1}$

  3 = 3D / 3×3 matrix

- All of the attitude parameterizations we talked about are representations of the $SO(3)$ group.

Quaternions Pt. 1:

- Quaternions are *the* standard in aerospace (and robotics and computer graphics) for simulation, control, and state estimation applications.

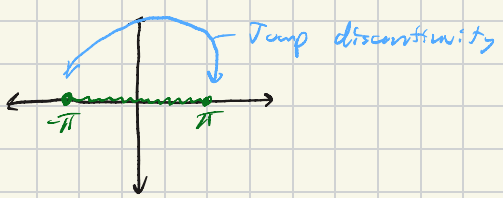
- They have a reputation for being confussing that isn't helped by the huge variance in notation and math conventions in the literature.

⇒ Never blindly trust quaternion math you see in books/papers/code. *Always* re-derive yourself.

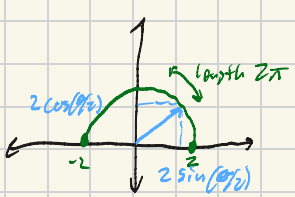
- We'll build this up starting with planar rotations to build intuition and develop good conventions and notation.

- Set of all planar rotations (1-DOF):

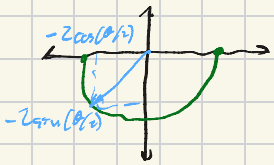
$$-\pi < \theta \leq \pi$$



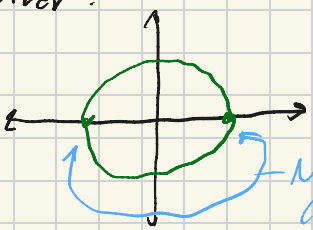
- "Pop Up"



- Copy + Reflect Through Origin:



- Glue copies together:



* To match standard conventions, normalize so vectors have unit lengths and rotate the picture 90° so $\theta = 0$ is at $q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

"planar quaternion"

* Note: $q(\theta) = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{bmatrix} \Rightarrow q(\theta + 2\pi) = \begin{bmatrix} \cos(\theta/2 + \pi) \\ \sin(\theta/2 + \pi) \end{bmatrix} = -q(\theta)$

$\Rightarrow$ q and $-q$ represent the same rotation

- We say that quaternions "double cover" the rotations

Combining Planar Rotations:

$$\theta_3 = \theta_1 + \theta_2$$

$$\Rightarrow \quad q_3 = \begin{bmatrix} \cos\left(\frac{\theta_1+\theta_2}{2}\right) \\ \sin\left(\frac{\theta_1+\theta_2}{2}\right) \end{bmatrix} = \left(\begin{bmatrix} \cos\left(\frac{\theta_1+\theta_2}{2}\right) & -\sin\left(\frac{\theta_1+\theta_2}{2}\right) \\ \sin\left(\frac{\theta_1+\theta_2}{2}\right) & \cos\left(\frac{\theta_1+\theta_2}{2}\right) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

- We can also recognize this as a 2D rotation matrix applied to $q(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\Rightarrow \quad q_3 = \underbrace{\begin{bmatrix} \cos(\theta_2/2) & -\sin(\theta_2/2) \\ \sin(\theta_2/2) & \cos(\theta_2/2) \end{bmatrix}}_{L(q_2)} \underbrace{\begin{bmatrix} \cos(\theta_1/2) \\ \sin(\theta_1/2) \end{bmatrix}}_{q(\theta_1)} = \begin{bmatrix} C_2 C_1 - S_2 S_1 \\ S_2 C_1 + C_2 S_1 \end{bmatrix}$$

- This is equivalent to complex multiplication:

$$(C_1 + S_1 i) * (C_2 + S_2 i) = (C_1 C_2 - S_1 S_2) + i(C_1 S_2 + C_2 S_1)$$

- We'll introduce the notation $L(q)$:

$$q_3 = L(q_2)\, q_1$$